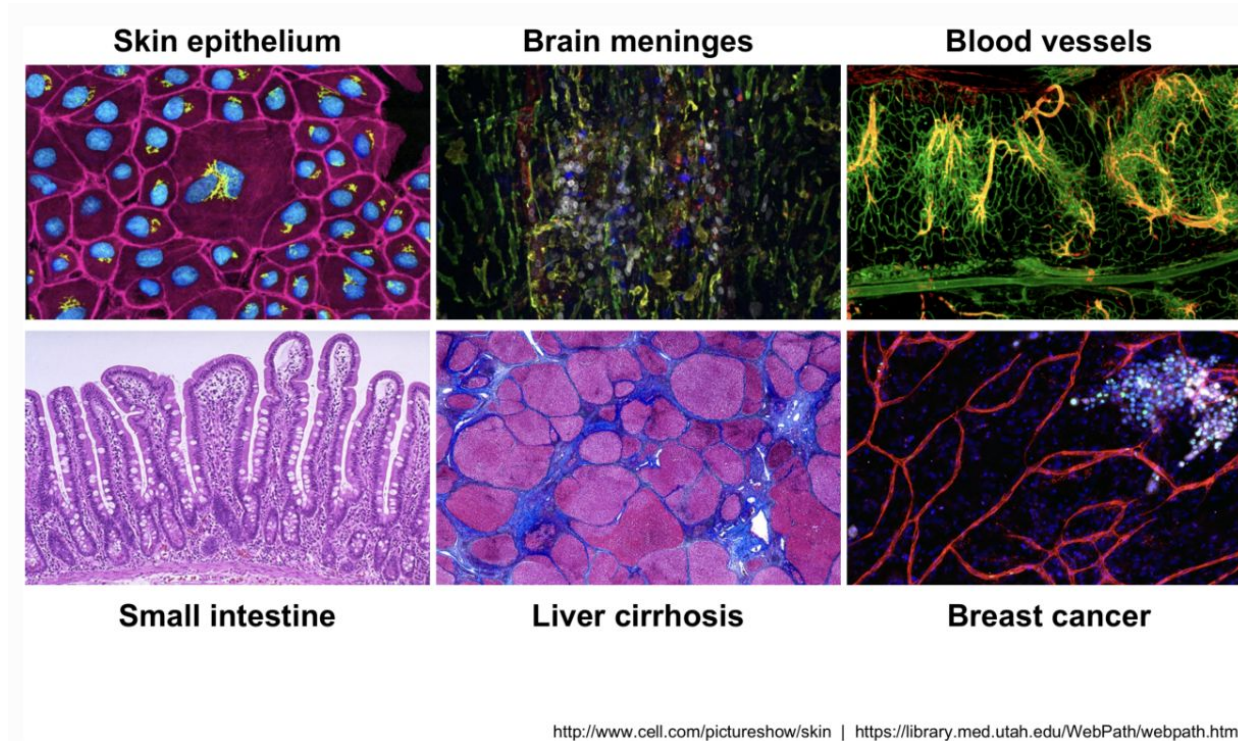


Day 4: SingleCell Analysis

Adapted from: <https://hbctraining.github.io/Intro-to-scRNAseq>

Why Single-cell RNA-seq?

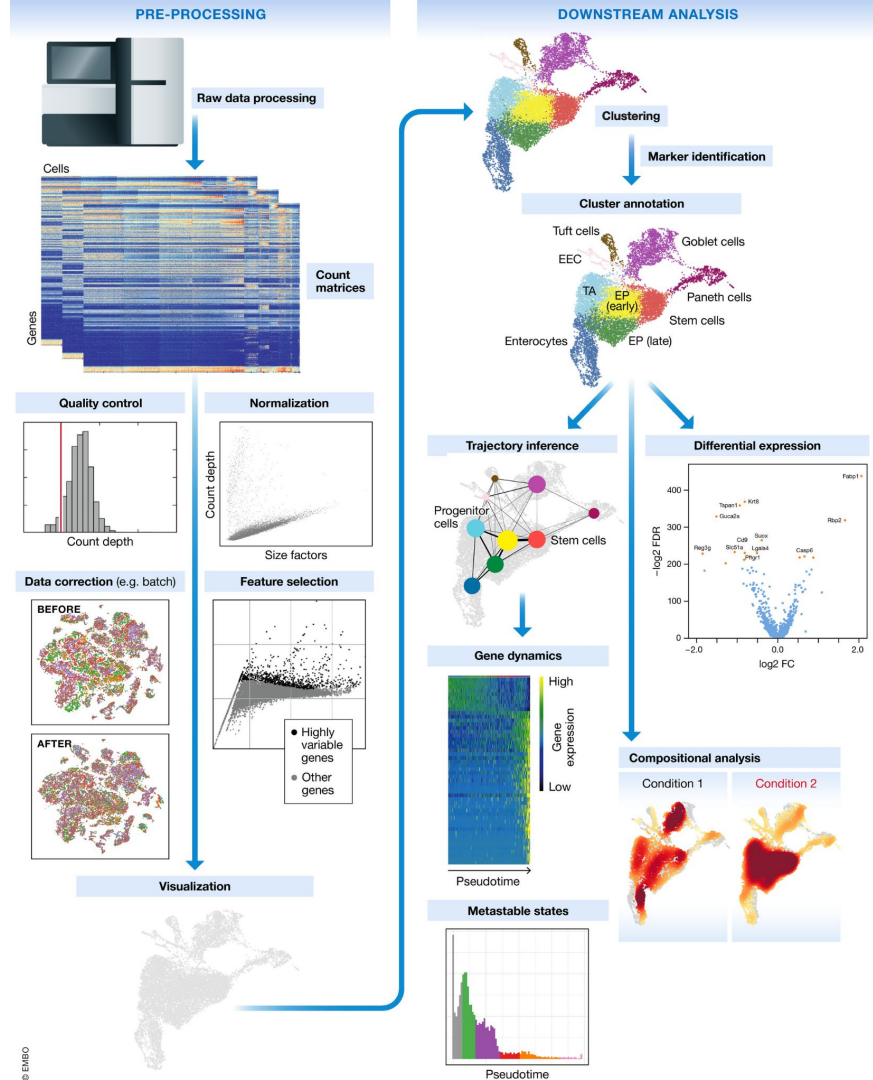


Why Single-cell RNA-seq?

- Explore which cell types are present in a tissue
- Identify unknown/rare cell types or states
- Elucidate the changes in gene expression across time or states
- Identify genes that are differentially expressed in particular cell types between conditions (e.g. treatment or disease)

From reads to UMAPs and beyond

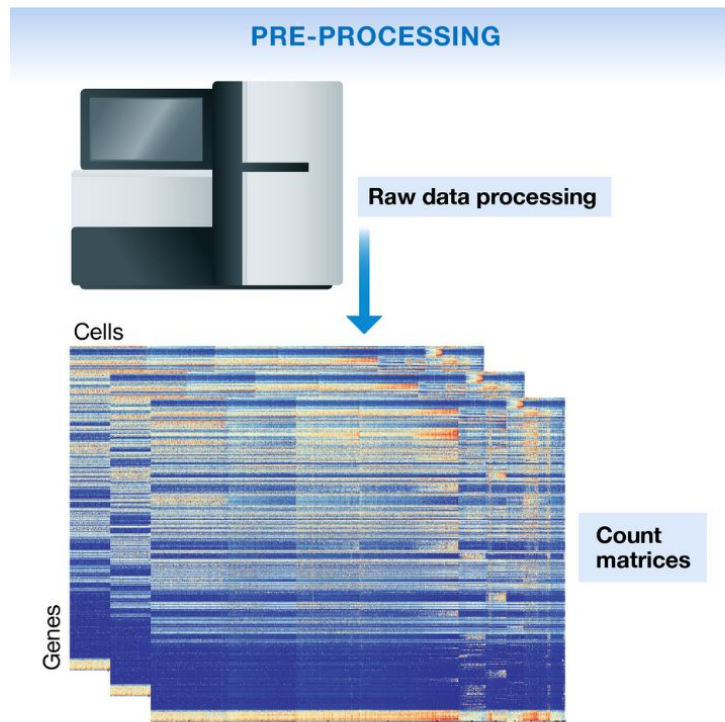
- Generation of the count matrix : forming reads, demultiplexing samples, mapping and quantification
- Filtering out poor quality cells
- Clustering cells based on similarities in transcriptional activity
- Marker identification and cluster annotation
- And more ...



Step 1: Alignment

Options:

- CellRanger
- STARsolo
- Alevin
- Kallisto



Step 1: Alignment

- Step 1.1: Build Genome Reference

```
mkdir -p cellranger_ref_main
cellranger mkref \
  --genome="hg38" \
  --fasta="hg38.p14.fa" \
  --genes="gencode.v49.annotation.gtf" \
  --nthreads=10 \
  --memgb=32 \
  --ref-version=1.0.0 \
  --output-dir=cellranger_ref_main
```

Step 1: Alignment

- Step 1.2: Alignment

```
cellranger count --id=sample345 \  
  --transcriptome=/path/to/reference/genome \  
  --fastqs=/path/to/fastq/files \  
  --sample=mysample \  
  --create-bam=true \  
  --localcores=8 \  
  --localmem=64
```

This will result in a directory structure like this:

```
E2_outs/  
├── barcodes.tsv -> filtered_feature_bc_matrix/barcodes.tsv  
├── cellsorted_possorted_genome_bam.bam  
├── features.tsv -> filtered_feature_bc_matrix/features.tsv  
├── filtered_feature_bc_matrix  
│   ├── barcodes.tsv  
│   ├── features.tsv  
│   └── matrix.mtx  
├── filtered_feature_bc_matrix.h5  
├── matrix.mtx -> filtered_feature_bc_matrix/matrix.mtx  
├── possorted_genome_bam_8T77P.loom  
├── possorted_genome_bam.bam  
├── possorted_genome_bam.bam.bai  
└── web_summary.html
```


10,985

Estimated Number of Cells

171,996

Mean Reads per Cell

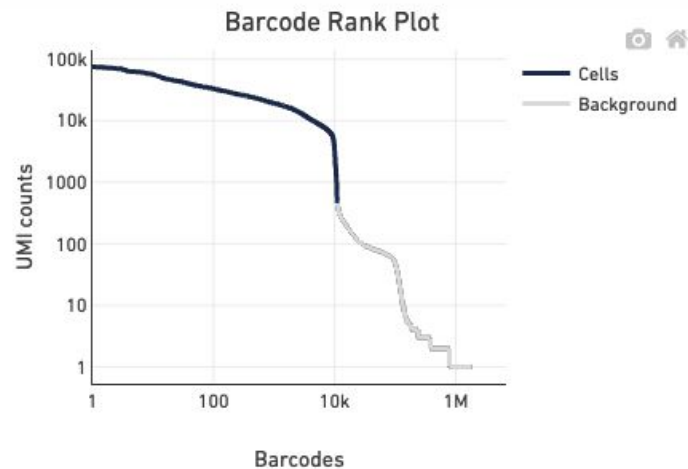
2,358

Median Genes per Cell

Sequencing ?

Number of Reads	1,889,370,721
Valid Barcodes	98.2%
Valid UMIs	99.9%
Sequencing Saturation	88.4%
Q30 Bases in Barcode	96.0%
Q30 Bases in RNA Read	93.3%
Q30 Bases in UMI	95.6%

Cells ?



Estimated Number of Cells	10,985
Fraction Reads in Cells	94.6%
Mean Reads per Cell	171,996
Median Genes per Cell	2,358
Total Genes Detected	26,061
Median UMI Counts per Cell	8,773

Mapping

Reads Mapped to Genome	97.3%
Reads Mapped Confidently to Genome	94.8%
Reads Mapped Confidently to Intergenic Regions	3.8%
Reads Mapped Confidently to Intronic Regions	29.4%
Reads Mapped Confidently to Exonic Regions	61.5%
Reads Mapped Confidently to Transcriptome	58.8%
Reads Mapped Antisense to Gene	1.5%

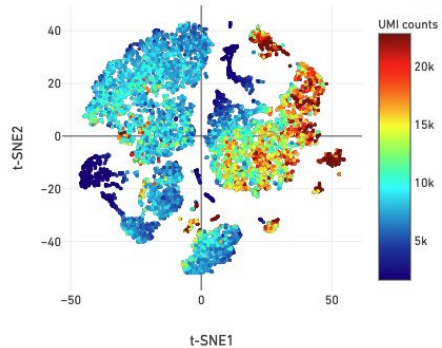
Sample

Sample ID	SC3_v3_NextGem_SI_PBMC_10K
Sample Description	10k Human PBMCs, Single Indexed
Chemistry	Single Cell 3' v3
Reference Path	...nger-4.0.0/refdata-gex-GRCh38-2020-A
Transcriptome	GRCh38-2020-A
Pipeline Version	cellranger-4.0.0

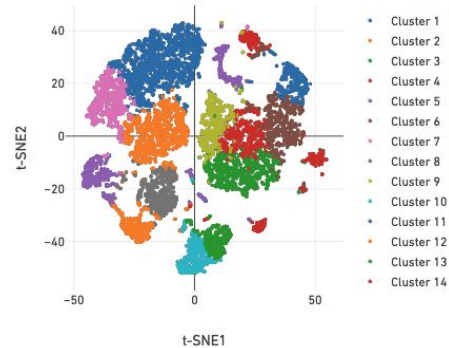
t-SNE Projection ?

Clustering Type: **Graph-based**

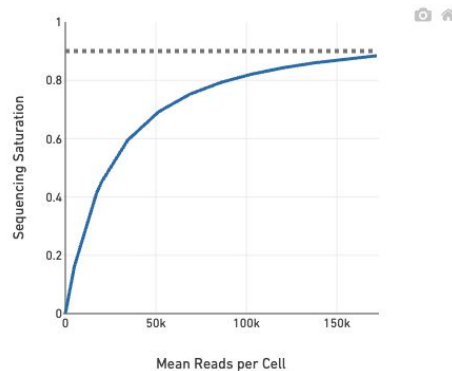
t-SNE Projection of Cells Colored by UMI Counts



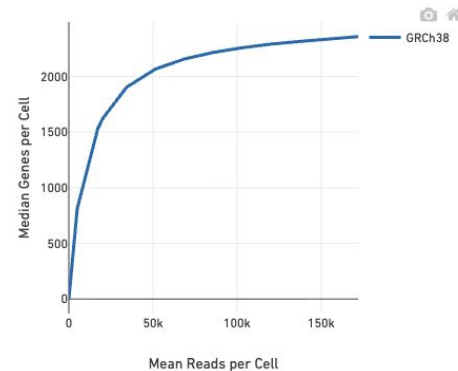
t-SNE Projection of Cells Colored by Clustering



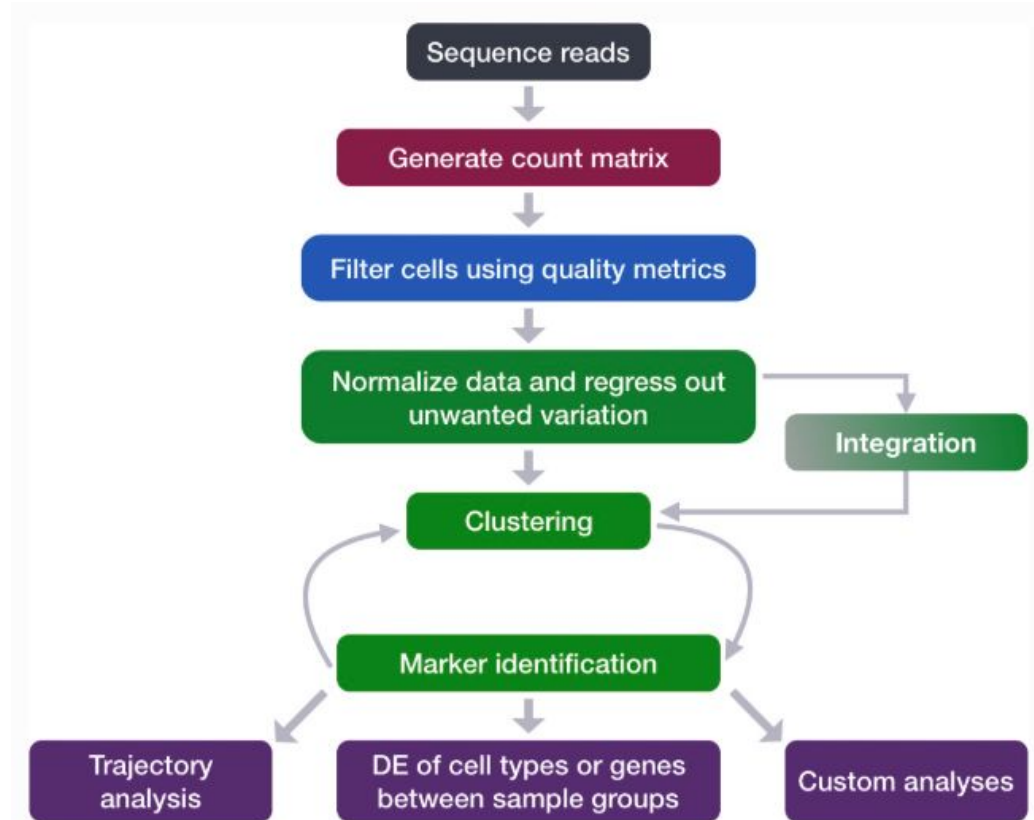
Sequencing Saturation ?



Median Genes per Cell ?



Bioinformatic Flow Overview



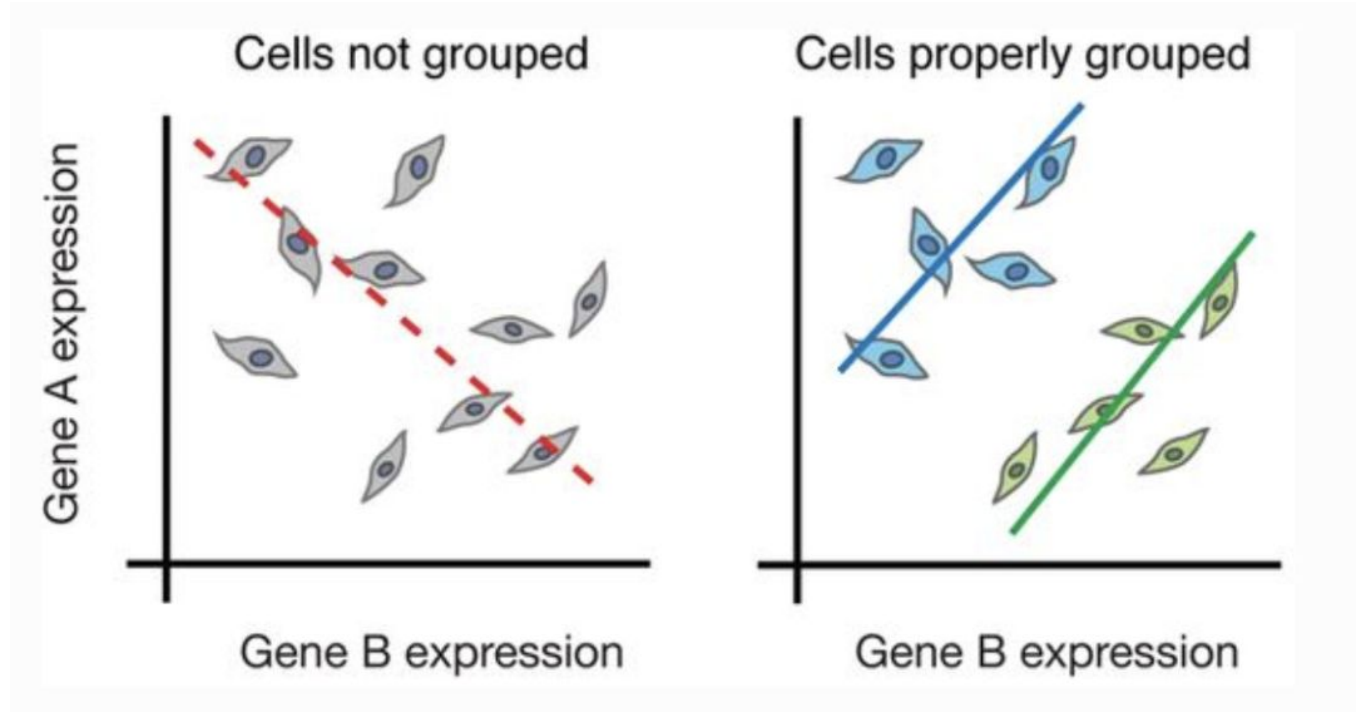
Now to our activity!

<https://simo-es-costa-lab.github.io/genomics-course-usp2026/scrnaseq/practical.html>

Day 5 | 02 Jul 2026

Time	Type	Location	Activity
09:00–10:30	Lecture	Queijinho A5	Lecture 07: Genome Architecture and Function (MSC)
11:00–12:00	Lecture	Queijinho A5	Lecture 08: Chromatin Organization (APA)
13:30–14:15	Science Talk	Multimedia Room (B1)	<i>Spatial Patterning of the Epigenome</i> (APA)
14:30–16:30	Bootcamp	Multimedia Room (B1)	Single-Cell RNA-seq Data Analysis (Helena Conceição)
16:30–18:00	Project	Multimedia Room (B1)	Project Development

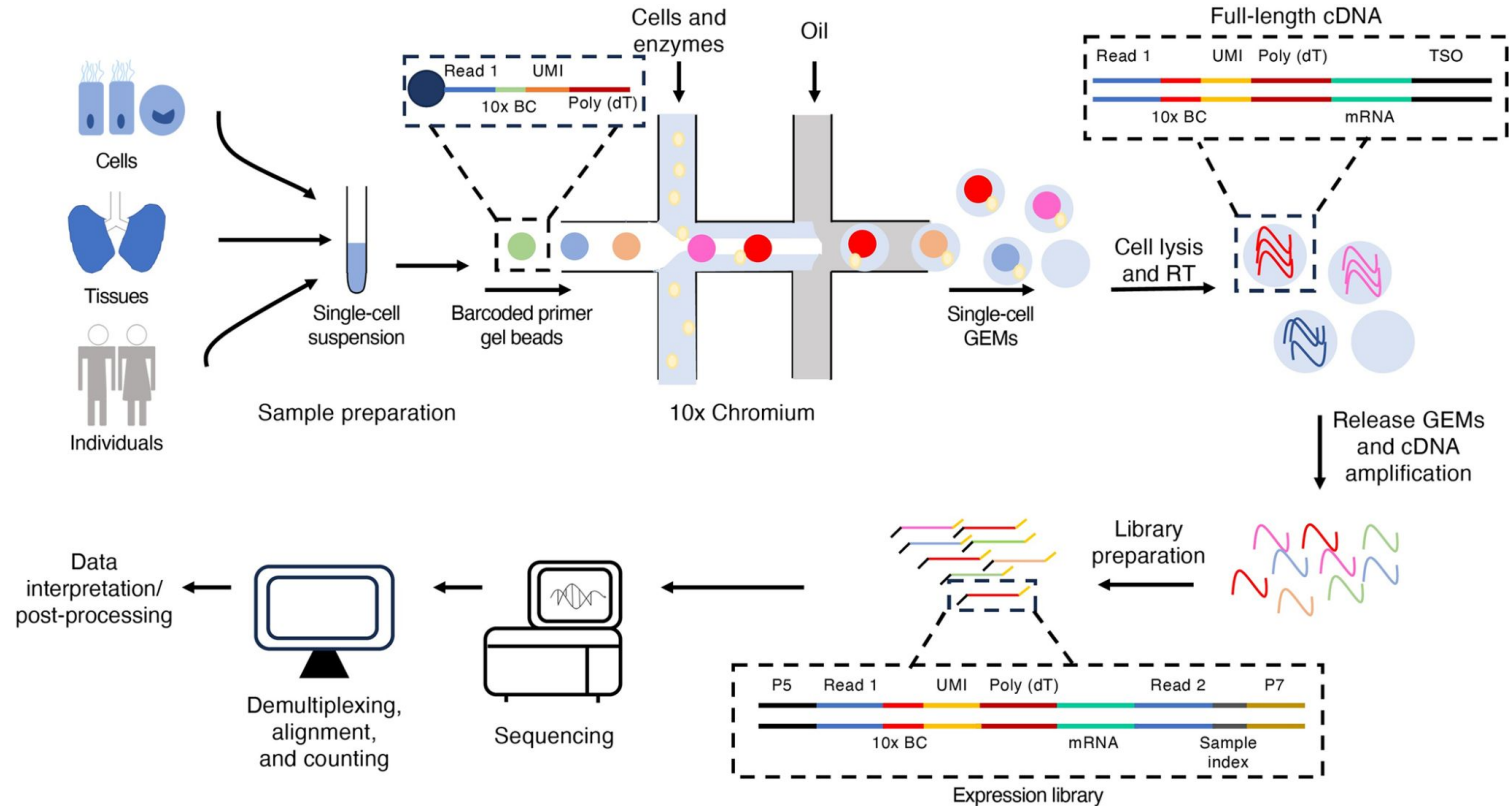
Bulk vs Single-cell



Set Backs

- Large volume of data
- Low depth of sequencing per cell
- Biological variability across cells/samples
- Varying rates of RNA processing
- Environmental stimuli
- Temporal changes
- Library quality
- Amplification bias
- Batch effects

Library prep and sequencing



Cell barcodes and UMI

